SWINBURNE UNIVERSITY OF TECHNOLOGY

COS30019: Introduction to Artificial Intelligence

Unit Review &

Exam Revision

1

Announcements

- **Final Assessment**:
  - When: Thursday 19th June 2025, 9am-12pm (**you must submit to Canvas by 12pm, 19th June 2025 AEST**)
  - **Online assessment; available on Canvas, OPEN BOOK**
  - **Note:** The deadline to submit is 12pm 19th June 2025 AEST even though I'll make the assessment available until 1pm 19th June 2025 AEST to accommodate late submissions (incurring late penalty)
- Your Unit. Your say. Survey
  - On Canvas
  - Please provide constructive feedback

2

AI/Definitions/Paradigms

- Define AI:
  - Different definitions
  - Can you briefly explain?
  - Can you compare between different AI paradigms?

SWIN BUR NE CENTRE FOR INFORMATION TECHNOLOGY RESEARCH

3

Intelligent Agents

- IA = AI systems that act rationally
- What does rationality mean?
- Performance measure?
- PEAS
- Task environment analysis?
- Agent structures?
  - Basic ones?
  - Advanced ones?

SWIN BUR NE CENTRE FOR INFORMATION TECHNOLOGY RESEARCH

4

AI Ethics and Responsible AI

- AI Ethics
  - What are current and potential impacts AI may have on humans and society?
  - Laws for AI/Robots?
- Emerging AI ethical issues:
  - Current issues presented by AI & AI-enabled technologies that require actions taken by AI designers/developers/users & regulators
- Responsible AI
  - Ethics (**in/by/for**) Designs
- Regulation guidelines & AI Trustworthiness Assurance

SWIN BUR NE CENTRE FOR INFORMATION TECHNOLOGY RESEARCH

5

Search-based problem solving agents

- Problem formulation?
- Search methods:
  - Uninformed/Blind
  - Informed/Heuristic
- Given a search problem, can you tell how different search methods will behave?
  - Make sure that you pay attention to Repeated State Check (RSC).

SWIN BUR NE CENTRE FOR INFORMATION TECHNOLOGY RESEARCH

6

Adversarial search/AI game playing

- Game tree
- Minimax
- Alpha-Beta pruning
- Expecti-minimax
- Given a game tree, can you:
  - Identify the To-Move player at each layer of the tree?
  - Compute the best move for the first player to move using MINIMAX??
  - Compute the best move for the first player to move using Alpha-Beta pruning algorithm??

SWIN BUR NE · CENTRE FOR INFORMATION TECHNOLOGY RESEARCH

7

Machine Learning - Key concepts

- "**Learning Problem**: A computer program is said to learn from experience E with respect to some task T and some performance measure P, if its performance on T, as measured by P, improves with experience E." - Tom Mitchell (1998)
- **Types of learning**:
  - Supervised (inductive) learning
  - Unsupervised learning
  - Semi-supervised learning
  - Reinforcement learning

SWIN BUR NE · CENTRE FOR INFORMATION TECHNOLOGY RESEARCH

8

Machine Learning - Key concepts

- **Supervised learning with Linear Regression**:
  - A statistical regression method used for predictive analysis
  - Computing the best-fit line: $y= h_{\beta}(x)=\beta_0+\beta_1 x$
  - Cost function (to measure the errors of the hypothesis h_{β}), e.g. MSE
  - Gradient descent – for optimization
- **Design a learning systems**:
  - A ML algorithm consisting of 3 major components: **Representation**, **Optimization**, and **Evaluation**

SWIN BUR NE CENTRE FOR INFORMATION TECHNOLOGY RESEARCH

9

Deep Learning - Key concepts

- Supervised learning with Logistic Regression
- Neurons:
  - A generalization of logistic regression
- Artificial Neural Network **(ANN)**
  - model = architecture + parameters
- Deep learning
  - Backpropagation algorithm for training a MLP

SWIN BUR NE CENTRE FOR INFORMATION TECHNOLOGY RESEARCH

10

Knowledge-based agents

- **Entailment (KB |= q)**
- **Models**
- **Truth table**
  - **Can you use it to show entailment???**
- **Validity/Satisfiability/Unsatisfiability**
  - **Valid? Unsatisfiable? Neither?**
- Forward chaining
- Backward chaining

SWIN BUR NE CENTRE FOR INFORMATION TECHNOLOGY RESEARCH

11

AI Planning

- Planning languages?
  - **STRIPS**, ADL, PDDL???
- **Formulate a planning problem (initial state, goals, action descriptions)**
- State-space search
  - Progression planning
  - Regression planning
- **Plan-space search**
  - **Partial Order Planning (POP)**
- **Can you use an AI planning technique to manually find a plan for a planning problem?**

SWIN BUR NE CENTRE FOR INFORMATION TECHNOLOGY RESEARCH

12

Probability/Reasoning with Uncertainty

- The problem of reasoning with uncertainty
- Probability from first principles
  - Basic axioms
  - Definitions (e.g., **conditional probability**, Independence, conditional independence, etc.)
  - **Conditioning**
- **Bayes rule**
  - **Can you use Bayes rule to perform inference with probability?**
  - **Can you answer questions from the tutorials/Practice Exam?**

SWIN BUR NE CENTRE FOR INFORMATION TECHNOLOGY RESEARCH

13

Probability - Key concepts

- Prior probability, e.g. $P(A_{90}) = 0.92$
- Conditional probability, e.g. $P(A_{90}|\text{accident on freeway}) = 0.74$
- $P(a) + P(\neg a) = 1$
- $P(a \mid b) + P(\neg a \mid b) = 1$
- Definition of Conditional Probability:

$$P(A \mid B) = P(A \wedge B)/P(B) = P(B \mid A)*P(A)/P(B)$$

- Conditioning:

$$P(A) = P(A \wedge B) + P(A \wedge \neg B) = P(A|B) * P(B) + P(A \mid \neg B) * P(\neg B)$$

SWIN BUR NE CENTRE FOR INFORMATION TECHNOLOGY RESEARCH

14

Probability - Key concepts

- Causal reasoning (using Bayes' rule):
  - Diagnostic reasoning from causal probability:
  - ***P(Cause | Effect) = (P(Effect | Cause) * P(Cause)) / P(Effect)***
- Examples:
  - Cause: Cavity / Effects: Xray, toothache
  - Cause: Disease / Effects: Symptoms (fevers, sore throat), test positive
  - Cause: Faulty alternator / Effects: Car won't start or frequently stalled
  - Cause: Bad credit applicant / Effects: AI system at the bank raises a warning on the application

15